

Phanthera: An Adaptive VR Exergame for Posture-Aware Recovery During Work Breaks

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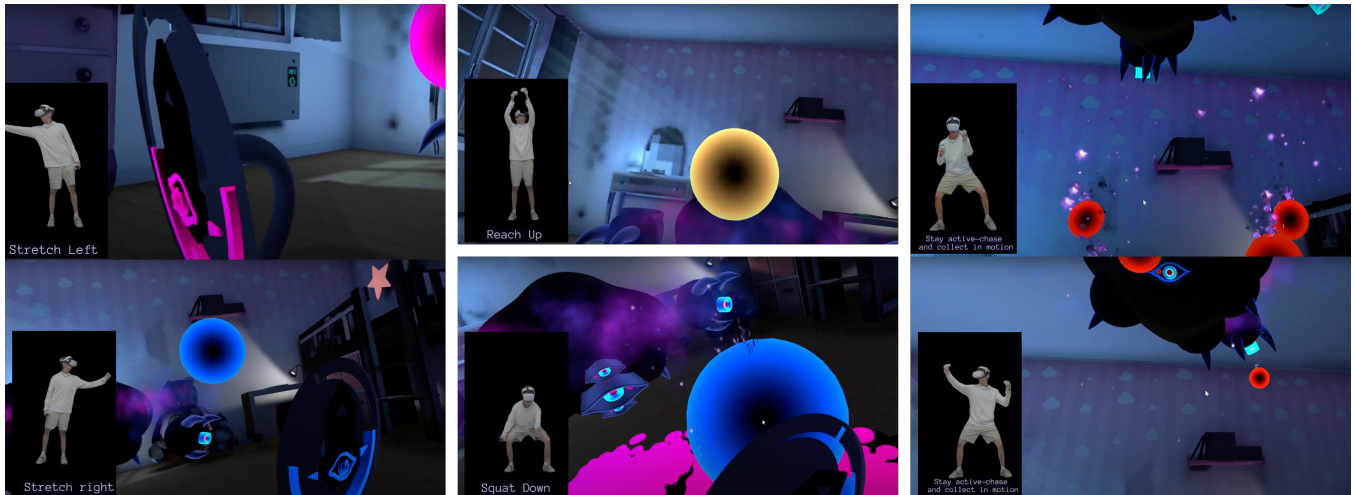


Figure 1: Phanthera, an adaptive VR exergame for posture-aware recovery. Players act as Dream Guardians, using guided stretches and full-body actions to intercept orbs. Sessions adapt to recent posture patterns to balance loading across muscle groups.

Abstract

Prolonged sedentary work and poor posture are linked to musculoskeletal strain and mental fatigue. While existing interventions

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encourage movement, they often overlook individual posture patterns and lack engaging user experiences. We present *Phanthera*, an adaptive VR exergame designed to support active recovery during work breaks. The system passively monitors user behavior (e.g., head tilt, limb dominance, inactivity) and uses this data to tailor ten-minute VR sessions targeting specific muscle groups. Players use color-coded “magic mirrors” to block incoming orbs through guided stretches, squats, and full-body movements. Gameplay adjusts in real time to promote balance, mobility, and periodic high-intensity engagement. By combining immersive interaction with posture-aware adaptation, Phanthera provides both physical relief and cognitive refreshment, offering a personalized approach to workstation wellness.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Ubiquitous and mobile computing design and evaluation methods*; *Interaction techniques*.

Keywords

virtual reality, exergames, posture, sedentary work, adaptive systems, workplace wellness

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1 Introduction and Background

Extended periods of fixed posture are common across a wide range of professions. The increasing reliance on laptops for work-related tasks has become a significant aspect of modern labor, as evidenced by recent data [12]. Studies indicate that employees may spend up to 75% of their work time seated at a computer [16]. While computers have undeniably facilitated a surge in productivity, their extensive use is not without adverse effects. Notably, there is a well-established link between the duration of computer use and an increased incidence of musculoskeletal symptoms [4, 10]. Over time, sustained postural demands can lead to muscular fatigue, reduced mobility, chronic discomfort, and elevated risk of injury [8].

Regular physical activity has been shown to reduce the risk of chronic conditions and, from an organizational standpoint, lower healthcare costs and employee absenteeism [7]. The World Health Organization recommends integrating more physical activity into the workday and allowing flexible, movement-friendly break schedules [17]. Effort have been put on investigating activity in workstation to enhance wellbeing. For example, Choi et al. explored the under desk elliptical trainers [5], and Haliburton et al. implemented a standing desk that raise the desk during breaks and was shown to have positive effect on user experience [9]. Ding et al. further demonstrated that standing and stretching breaks were more effective in alleviating fatigue than passive breaks or simple standing and walking [6]. Furthermore, Some efforts have targeted specific body regions; for example, Mihajlovic et al. developed a serious game focused on rehabilitative neck movements [13].

Despite this, conventional break strategies often fail to address the accumulated postural imbalances that result from prolonged work, nor do they consistently promote cognitive disengagement. To address these limitations, prior research has introduced exergames designed to promote physical activity as a more engaging form of work break intervention [3, 15]. Recent studies have shown that XR-based exergames can further enhance engagement and motivation by combining physical activity with immersive, playful experiences [1, 2, 11, 14, 18]. However, most existing systems do not account for the user's posture or movement behavior accumulated throughout the workday, nor do they adapt training to target specific areas of muscular fatigue or postural strain.

Hence, we present Phanthera (Figure 1), an adaptive VR motion-based game designed to promote active recovery through personalized, posture-aware training. By integrating real-time behavioral data and immersive XR interaction, Phanthera creates short, engaging sessions that both activate the body and offer psychological refreshment. The system dynamically adjusts gameplay based on user posture and movement history, enabling targeted interventions that align with the user's actual physical needs. We demonstrate how Phanthera personalizes the break experience to meet the evolving physical needs of the user through the development of several proof-of-concept scenarios, such as promoting lateral stretches for asymmetrical posture, encouraging squats after prolonged sitting, and activating intensity mode during periods of cognitive fatigue. This work highlights the potential of adaptive VR exergames to serve as effective, engaging tools for supporting physical wellbeing and mental recovery in sedentary work environments.

2 PhanThera

2.1 Design Consideration

Phanthera explores the design considerations at the intersection of adaptive gameplay, embodied interaction, and emotion regulation for workers. The game is purpose-built for short, restorative breaks during the workday, emphasizing both physical activation and emotional decompression through engaging VR experiences.

Personalized Physical Engagement Through Behavioral Adaptation. Phanthera's real-time adaptivity is driven by behavioral data collected during work hours. Unlike fixed-exercise VR systems, Phanthera continuously analyzes user posture, hand use, and activity patterns to tailor the in-game physical challenges. This creates a closed feedback loop where in-game movement compensates for real-world imbalances. For example, if a player exhibits prolonged static posture with a downward gaze, the game increases the frequency and height of incoming projectiles, encouraging upper-body extension and neck mobility.

Immersive Game Design for Mental Refreshment and Enjoyment. To maximize the relaxation of short play sessions, Phanthera adopts VR to create a high-immersion game design to fully isolate players from their work context. The VR dreamscapes are vivid and stylized, with ambient audio and responsive environments that create a strong sense of presence. This immersion is key to psychological distance, offering a clear mental boundary between the stressors of work and the rejuvenating space of play. Combined with meaningful player agency and challenge, these elements promote enjoyment, reduce cognitive fatigue, and provide a refreshing reset during work.

Integration into Work Rhythms and Microbreaks. Previous work has shown that microbreaks during work could enhance worker wellbeing and productivity [3, 6]. Phanthera is explicitly designed for integration into daily work schedules, aligning with the concept of microbreaks. Game sessions are short and impactful, offering both physical reset and cognitive detachment from desk tasks. This situates the game in a unique design niche: not just exercise or entertainment, but restorative interludes that enhance workplace wellbeing.

2.2 System Architecture

The Phanthera system architecture is composed of three modules (Figure 2) designed to provide adaptive and immersive experiences: 1) the behavioral analysis module, 2) the adaptive gameplay parameter, and 3) the immersive VR game environment.

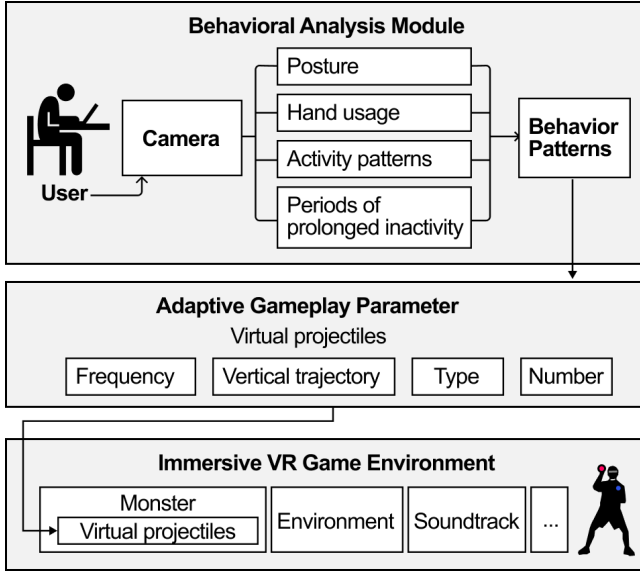


Figure 2: System Architecture: The behavioral analysis module, the adaptive gameplay parameter, and the immersive VR game environment

2.2.1 Behavioral Analysis Module. This module captures and analyzes behavioral data from users throughout their workday. Data is collected continuously using the webcam (or phone camera). During regular computer-based work sessions, Phanthera passively monitors upper-body posture using MediaPipe. From these, it computes real-time metrics such as head pitch/yaw, neck tilt, and shoulder alignment. Sampling occurs at 1 Hz, and data is stored in time-stamped JSON entries for hourly evaluation.

To quantify the physical strain or imbalance of each joint, we define a **Need Score (NS)** as:

$$NS_j = \sum_{t=1}^T \left(\left| P_j(t) - P_j^{\text{neutral}} \right| \times D_t \right) \quad (1)$$

where:

- NS_j : Need Score for joint j
- $P_j(t)$: Postural angle or position of joint j at time t
- P_j^{neutral} : Ideal or neutral posture value for joint j
- D_t : Duration weight for the posture bias at time t
- T : Number of time samples in the evaluation window (typically 3600 per hour)

This score captures both the *degree* and *duration* of deviation from neutral posture. A higher score indicates more significant or prolonged postural strain, triggering personalized intervention in gameplay.

At the beginning of each VR session, the system ranks joints based on their computed Need Scores. This ranking drives two adaptive mechanisms in gameplay:

- (1) **Orb Directional Bias:** Orbs are launched more frequently toward the region with the highest need score. For example, if the left shoulder has the highest NS, the system increases the frequency of red orbs targeting the left hand, encouraging corrective movement.
- (2) **Movement Type Emphasis:** Orb types and trajectories are adjusted to stimulate motions that address specific imbalances. High head tilt scores trigger more yellow overhead orbs, prompting upright posture and full arm extension.

This approach allows *Phanthera* to personalize gameplay as a form of embedded physical intervention, gradually promoting ergonomic balance while maintaining an engaging game experience.

2.2.2 Adaptive Gameplay Parameter. The adaptive gameplay parameter dynamically modifies the game challenges based on data from the *behavioral analysis module*, by dynamically adjusting game parameters. This module tailors physical interactions to address identified user behavior status and special needs. For instance, detecting a prolonged static posture triggers adaptive parameter adjustments to compel the player to perform upper-body extension and neck mobility exercises, such as increasing the frequency and vertical trajectory of incoming virtual projectiles.

2.2.3 Immersive VR Game Environment. This module provides a visually rich and engaging space distinct from the users' work setting. It leverages immersive soundscapes and responsive environments to maximize psychological disengagement and refreshment. The environment supports users' body training by creating a clear mental boundary between work stressors and the rejuvenating gameplay experience. In the VR game environment, users interact with compelling dreamscapes designed explicitly to facilitate mental relaxation and physical rejuvenation within short, impactful sessions.

To ensure smooth transitions and continuous adaptation during play, these components are seamlessly integrated. The *Behavioral Analysis Module* provides constant feedback to the *Adaptive Gameplay Parameter*, enabling personalised game settings. Simultaneously, the *Immersive VR Game Environment* sustains engagement and encourages regular participation by providing enjoyable, meaningful interactions tailored precisely to user needs identified during their daily routines.

2.3 Workflow and Game Design

Phanthera is designed to support knowledge workers through adaptive and immersive VR experiences embedded into their daily routines. Each session is part of an ongoing feedback loop that alternates between passive monitoring and gameplay, with the goal of reducing physical strain and promoting emotional recovery.

2.3.1 Phase 1: Passive Monitoring During Work. During normal computer-based work, Phanthera passively tracking behavioral data through the camera, recording the posture (e.g. Head angle, neck tilt, shoulder alignment). Every hour, the system evaluates recent behavior and determines the break plan. It issues a reminder from the monster, such as "Guess who's back? You better show up

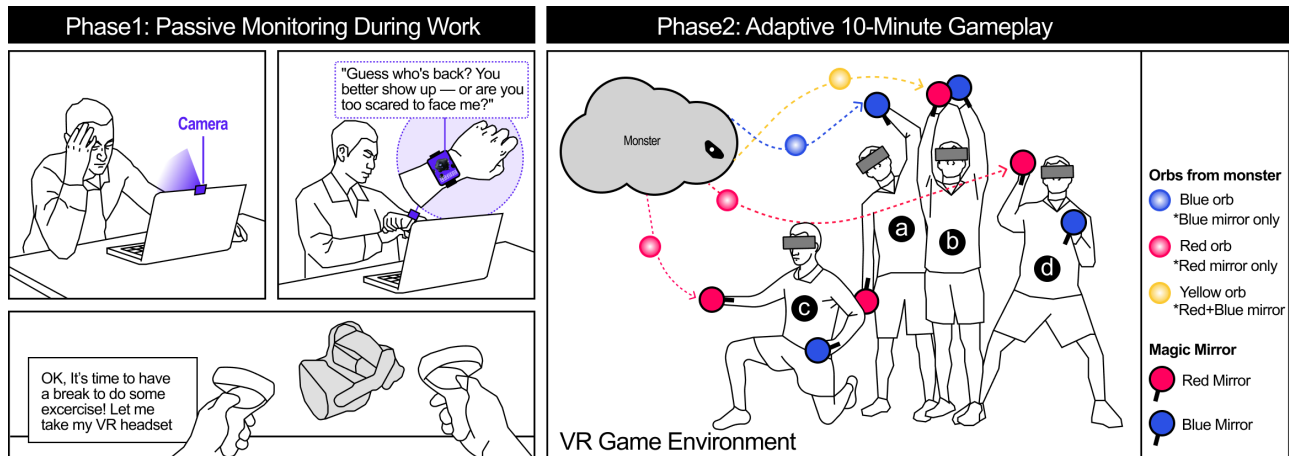


Figure 3: Phanthera operates in two phases. Phase 1: Passive Monitoring During Work. A camera deployed on the user’s computer passively tracks behavioral data. Every hour, the system evaluates recent activity to determine a personalized break plan and sends a reminder. Phase 2: Adaptive 10-Minute Gameplay. During the break, the user engages in a 10-minute game, using physical movements to control red and blue magic mirrors to block incoming attacks from a monster. The differences in the orbs prompt different body movements, achieving targeted physical training goals.

— or are you too scared to face me?" This passive phase ensures low-friction integration into the workday while collecting actionable insights that drive gameplay adaptation.

2.3.2 Phase 2: Adaptive 10-Minute Gameplay. In the VR session, the user becomes a Dream Guardian equipped with two magic mirrors: red (left hand) and blue (right hand). A monster in front of the player continuously fires orbs to corrupt the dream world. To defend the dream, the player must match the color of incoming orbs with the correct mirror and reflect them at the monster to deal damage. The goal is to minimize dream contamination while fighting back. There are two victory conditions: 1) Reduce the monster's HP to zero; 2) Prevent the corruption of the dream from reaching 100%.

The gameplay is carefully choreographed to promote targeted movement. There will be two mode, stretching mode and intensity mode.

In stretching mode, players can specifically target and train certain parts of their body.

- Red orb (blocked with left hand): Flies to the left, stretching their left arm; Flies to the right, encouraging the user to stretch their left arm across the body (Figure 3 a).
- Blue orb (blocked with right hand): Flies to the right, stretching their right arm; Flies to the left, encouraging the user to stretch their left arm across the body (Figure 3 a).
- Yellow orb: Requires the player to stack both mirrors together and raise them overhead, promoting full arm extension (Figure 3 b).
- Low-flying orbs: Appear in all colors and target lower positions, requiring the player to squat and rise, engaging the legs and core (Figure 3 c).

In the intensity mode (Figure 3 d), training goes beyond targeted exercises and focuses on full-body relaxation. Additionally, the monster increases orb speed and frequency for short bursts.

This mode raises the player's heart rate and enhances dynamic movement.

3 Scenarios

3.1 Envision Scenarios

To illustrate how **Phanthera** adapts to users' real-world behaviors, we implemented a set of proof-of-concept services combining passive posture monitoring with real-time gameplay adaptation. A standard webcam captures user movement, while an analysis engine detects patterns in behavior and activity levels. Based on these observations, the system dynamically configures short VR sessions that promote targeted physical engagement. Below, we present three scenarios highlighting personalized interventions:

- **Mary — Upper Body Realignment.** A desk-bound strategist with postural asymmetry (leaning left, tilting head, neck tension) is guided through a session emphasizing vertical and lateral arm stretches. Gameplay encourages shoulder elevation and cross-body reaching to counteract her habitual positioning.
- **Alex — Lower Body Re-engagement.** A sedentary programmer with minimal leg movement is prompted to perform repeated squats and upright motions. The session targets circulation and mobility restoration through low-ground orb deflection challenges that activate the legs and core.
- **Lina — Fatigue Recovery.** A fatigued project manager exhibiting slouching and drowsiness is engaged in a high-intensity, full-body session. Orbs appear at varied positions and speeds, requiring alertness, quick squats, overhead reaches, and lateral hand coordination—delivering both cognitive stimulation and light cardiovascular activity.

These personalized gameplay adjustments demonstrate Phanthera's ability to promote physical well-being through contextual adaptation during the workday.

3.2 Pilot User Study

We conducted a pilot user study to initially evaluate the system's effectiveness, engagement, and areas for improvement. Four knowledge workers (two male and two female) were recruited, all of whom were engaged in research-related occupations that required prolonged periods of sedentary computer work. With participants' consent, we installed a camera in front of each participant's workstation. They were instructed to carry out their tasks at a normal working pace. Every hour, participants received a prompt from the researchers inviting them to engage in a 10-minute gameplay session. Each participant completed three gameplay sessions in total. Upon completion of the experiment, we conducted semi-structured interviews to gather their feedback on the system.

The results indicated that most participants found the system engaging and novel. They described the experience as "*playful*" and "*motivating*," and expressed willingness to continue using the system in the long term. A frequently mentioned observation was the need to look up frequently to track incoming orbs, which participants noted "*naturally encouraged them to lift their heads and move more*" without being disruptive. When asked whether they noticed the game's adaptive content, two participants reported perceiving a difference in gameplay settings. One noted that "*the balls tended to fly toward one side*," which prompted her to move that side of her body more actively. Lastly, participants offered further design suggestions, such as "*adding lower-body exercises*" and "*deploying the system in a mixed reality (MR) environment to better perceive the surrounding space*."

Overall, the system was perceived as a low-friction way to introduce meaningful physical activity into sedentary work routines, with game elements significantly enhancing adherence and enjoyment.

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